

Reply to MERIS_PROPOSAL_3SLOT.pdf / 三槽轮换提案回复

Date: 2026-09-24 To: R2 Engineering (Y. Hirata), AZ-27SS e-paper garment project From: LIN Firmware: FW_260923 (protocol V1.4)

Overall / 总体结论

We checked your 3-slot rotation (17 → 18 → 19) against FW_260923. It works with the existing firmware; no firmware change is needed. Each picture gets a write window of about two intervals (≈ 16 s), and 36 boards of 0x13 take 8–9 s, so there is roughly 7 s of margin. The sequence is safe for the controller and for the stored data. The remaining point is the paper-quality risk of 1 s intervals, which you accept in section 1 of your proposal.

我们对照 FW_260923 核查了你们的三槽轮换方案 (17 → 18 → 19)：现有固件即可支持，无需改动固件。每张画的写入窗口约为两个间隔 (≈ 16 s)，36 块板的 0x13 需 8–9 s，裕量约 7 s。该流程对控制器和数据都是安全的；剩下的只有 1 s 间隔的纸面质量风险，如你们提案第 1 节所述由贵方承担。

Q1. Slots 17 and 18 / 槽位 17 和 18

Yes, slots 17 and 18 can be overwritten permanently (0x13 / 0x1F / 0x1B). The factory pictures in them will be lost, which you already accept. In the firmware all 20 slots (0–19) are the same: same storage layout (colour data + per-segment delay table + slot configuration), same capacity, same behaviour. No slot is reserved, and there is no difference between slots 0–18 and slot 19. The only autoplay-related note: the master board (address 0x01) auto-plays slot 0 once at power-up (slaves do not autoplay), so send one broadcast 0x17 after power-up (see Q4).

可以，槽位 17、18 可以永久覆盖 (0x13 / 0x1F / 0x1B)，其中的出厂图片会丢失，这点贵方已接受。固件里 20 个槽位 (0–19) 完全相同：存储结构 (颜色数据 + 逐段延迟表 + 槽位配置)、容量、行为都一样，没有保留槽位，槽位 0–18 与 19 之间没有差异。唯一与自动播放有关的注意点：主机板 (地址 0x01) 上电会自播一次槽位 0 (从机不自播)，上电后广播一次 0x17 即可 (见 Q4)。

Q2. Commands received during a refresh / 刷新过程中收到命令

(a) The ACK is returned immediately, also while the board is refreshing. (b) The command is executed immediately, not queued. FW_260923 receives and processes bus commands in parallel with the refresh. For 0x13, the storage write happens as soon as the last frame for that board arrives, and the ACK is sent after the write completes, so your measured 0.22–0.25 s per board already includes the erase/program time. (c) The receive buffer holds 16 frames for burst absorption. Since commands execute immediately, your flow (one frame per board, ACK awaited) will not hit this limit, and there is no "queue until the refresh ends" behaviour. Your 2026-08-14 measurement of queuing matches the older, pre-FW_260923 firmware; on FW_260923 execution is immediate.

(a) ACK 立即返回，板卡正在刷新时同样如此。(b) 命令立即执行，不排队。FW_260923 在刷新期间并行接收并处理总线命令。0x13 在该板最后一帧收齐时即执行存储写入，ACK 在写入完成后才发出，所以你们实测的每板 0.22–0.25 s 已包含擦写时间。(c) 接收缓冲为 16 帧，用于吸收突发。命令是即时执行的，你们的流

程（每板一帧、等 ACK）不会触及该上限，不存在“排队到刷新结束”的行为。你们 2026-08-14 测到的排队现象对应 FW_260923 之前的旧版固件；FW_260923 上是即时执行的。

Q3. Writing a slot that is not being displayed / 写入未在显示的槽位

Safe. The board latches the whole picture into internal RAM before the refresh starts, and the refresh itself never reads the slot storage back. Writing another slot S2 with 0x13 during the refresh of S1 cannot affect the picture being refreshed; the timing is not restricted in any way, mid-repaint included. For the opposite case you asked about: even if 0x13 arrives for the same slot that is currently refreshing, the image is already latched, so the ongoing refresh is unaffected and cannot tear. The new data takes effect the next time that slot is shown (the next 0x1D for that slot). The display interface and the storage chip also sit on two independent buses.

安全。刷新开始前，板卡已把整幅画面锁存进内部 RAM，刷新过程不会回读槽位存储。在对槽位 S1 刷新期间用 0x13 写另一槽位 S2，不会影响正在刷新的画面，时间上没有任何限制，刷新进行中也可以。你们问的反向情况：即使 0x13 写的是正在刷新的同一槽位，画面也已锁存，正在进行的刷新不受影响，不会撕裂；新数据在该槽下一次显示（下一发对该槽的 0x1D）时才生效。显示接口与存储芯片也在两条相互独立的总线上。

Q4. 0x17 stop before each write / 每次写入前的 0x17 停止

Not necessary. 0x13 is a pure storage command; it never starts or affects playback. Autoplay happens only once, at power-up (master board, slot 0). Recommended practice: after each system power-up, send one broadcast 0x17, then use only 0x1D per picture for the rest of the show. If 0x17 arrives while a refresh is in progress, it stops the playback state machine (no next slot, no completion notification), but it does not abort the physical refresh; the current refresh always runs to completion. No need to repeat it.

不需要。0x13 是纯存储命令，不会启动播放，也不影响播放。自动播放只发生一次，即上电时刻（主机板、槽位 0）。推荐做法：每次系统上电后广播一次 0x17，之后整场演出只按画面发 0x1D。若 0x17 在刷新进行中到达，它会停止播放状态机（不切下一槽、不发完成通知），但不会中止物理刷新，当前刷新总会完整跑完，无需重复发送。

Q5. 0x1D while a board is still refreshing / 板卡仍在刷新时收到 0x1D

It is not dropped. The board finishes the current refresh, then loads the newly commanded slot and refreshes it, so the board does one extra refresh, the same behaviour you observed on 2026-08-14. There is no firmware-side minimum interval between the end of one refresh and the next 0x1D; 1 s is fine for the controller (the paper-quality risk is on your side, as stated). Boards differ by only a few hundred ms, so a board that is still finishing when the next 0x1D arrives simply runs the two refreshes back to back, which matches what the show director wants for that board.

不会丢弃。板卡先完成当前刷新，随后装载新命令的槽位并刷新，即补刷一次，与你们 2026-08-14 观察到的行为一致。固件对“上一刷新结束到下一 0x1D”没有最小间隔要求；对控制器而言 1 s 没有问题（纸面质量风险由贵方承担，如前所述）。板间只差几百毫秒，仍在收尾的板收到下一发 0x1D 时会背靠背完成两次刷新，这也正符合演出导演对该板的期望。

Q6. 0x1E "switch to next slot" / 0x1E"切换下一槽位"

0x1E does not help this rotation, and please do not probe it on the bus. 0x1E is an internal coordination command that the master uses during 0x16 loop playback to advance slaves to the next slot; it has no "pre-

select the next slot" function. A board in slave mode that receives 0x1E will immediately display the slot given in data[0] and reply ACK_SUCCESS, so a probe would trigger a real, visible display during the show. In other modes the command is silently ignored (no reply). The rotation does not need it: the broadcast 0x1D already carries the slot number and starts all boards together.

0x1E 对该轮方案没有帮助，并且请勿在总线上探测该命令。0x1E 是 0x16 循环播放时主机推进从机切槽的内部协调命令，没有“预选下一槽位”的功能。从机模式的板收到 0x1E 会立即显示 data[0] 所指槽位并回复 ACK_SUCCESS，探测会在演出中触发真实的可见显示；其他模式下该命令被静默忽略（无应答）。轮转方案不需要它：广播 0x1D 自带槽位号，全板同时起刷。

Q7. Refresh duration / 刷新时长

Nominal ≈ 7 s from 0x1D to the finished picture. The duration is set by the display waveform and does not depend on the number of boards (after the broadcast each board refreshes its own panel in parallel), nor in practice on how many segments change or which colours are involved: every refresh is a full repaint, and fill ("no refresh") segments are skipped without changing the cadence. A few hundred ms of board-to-board variation is normal (panel busy timing). The only case where the duration grows by design is a slot with a 0x1F per-segment delay table ("sweep" picture): session $\approx$ max delay of the table + waveform tail. Please verify with your real 16-colour designs as planned in your section 5.

标称约 7 s（从 0x1D 到画面完成）。时长由显示波形决定，与板上数量无关（广播后各板并行刷新自己的屏），实际也与变化段数、涉及颜色基本无关：每次都是整幅重刷，填充（“不刷新”）段被跳过但不改变节奏。板间几百毫秒的差异属正常（屏体忙信号时序）。唯一会按设计变长的情况是该槽带有 0x1F 逐段延迟表（“扫描”画面）：场次 $\approx$ 表内最大延迟 + 波形尾巴。请按你们第 5 节的计划用真实 16 色设计目测验证。

Q8. Flash endurance / Flash 寿命

500–1000 writes per slot is well within the endurance of the storage chip (NOR flash, typically $\geq 100k$ erase/program cycles per sector), roughly a hundred times of margin, so wear is not a concern for a show and its rehearsals. A RAM-only "next picture" buffer does not exist in the firmware and is not needed for this rotation: the 3-slot scheme already solves the timing, and keeping the pictures in non-volatile storage means they also survive a board reboot mid-show.

每槽 500–1000 次写入远在存储芯片（NOR Flash，典型每扇区 ≥ 10 万次擦写循环）的寿命范围之内，裕量约一百倍，对一场演出及其排练没有磨损顾虑。固件没有“仅 RAM 的下一张缓冲”，该轮转方案也不需要：三槽方案已解决时序问题，且画面保存在非易失存储中，演出中途板卡重启画面也不丢失。

Q9. Power / 电源

No controller-side current or timing constraint. In particular there is no window during a refresh in which storage writes are forbidden (no "do not write within the first N ms" rule). The display interface and the storage chip use separate buses, and the storage write current (mA level) is negligible against the panel refresh current. The only timing cost of a write is the ≈ 0.2 s erase/program + ACK per board, already inside your 8–9 s budget for 36 boards. Measuring the 12 V rail at the last board during a back-to-back run, as you plan in section 5, will confirm this at system level.

控制器侧没有电流或时序方面的约束，特别是不存在刷新期间禁止写存储的时间窗（没有“开始后 N ms 内不要写”的规则）。显示接口与存储芯片使用相互独立的总线，存储写入电流（mA 级）相对屏体刷新电流可忽略。

写入的唯一时间代价是每板约 0.2 s 的擦写 + ACK，已包含在你们 36 块板 8–9 s 的预算内。按你们第 5 节的计划在背靠背连播时测末端板 12 V 轨，即可在系统层面确认这一点。

Additional notes / 补充建议

1. After power-up, and after any reboot of the master board, make the first command a single broadcast 0x17. A slave that reboots mid-show needs no special handling: it does not autoplay, and the next broadcast 0x1D brings it back in sync.
2. 0x1B / 0x13 / 0x1F are all persistent; re-sending after a reboot is only needed when you want to change values. Your practice ("once per slot, or after a board reboot") is harmless, so keeping it is fine.
3. The 0x1D data body only needs 1 byte (slot number). The optional second byte (interval) has no effect in single-shot mode.
4. Count the 0x13 ACKs per picture and re-send to boards that did not answer before firing the 0x1D, so no board is left showing a stale picture.
5. As a mitigation for the 1 s cadence: full-white refresh pictures during rehearsals and intermissions help the paper recover contrast. During the show itself, 1 s is safe for the controller and the data.

补充建议：

1. 上电后以及主机板每次重启后，第一条命令建议为广播 0x17（仅一次）。演出中途重启的从机无需特殊处理：从机不自播，下一发广播 0x1D 会自动将其拉回同步。
2. 0x1B / 0x13 / 0x1F 均持久化存储，重启后仅在需要修改数值时才须重发。你们“每槽一次、或重启后重发”的做法无害，可以保留。
3. 0x1D 数据体只需 1 字节（槽位号），可选的第二个字节（间隔）在单张模式下无作用。
4. 每张画统计 0x13 的 ACK 数，对未应答的板先补发、再发 0x1D，避免个别板显示旧图。
5. 作为 1 s 节奏的缓解措施：排练和幕间插入全白刷新画面，帮助纸面恢复对比度。正式演出期间，1 s 间隔对控制器和数据侧是安全的。

If anything you observe differs from what is described above, please tell me promptly; I will prepare a countermeasure as soon as possible.

如果贵方实测现象与上述描述有差异，请及时告知我，我会尽快给出尽可能的处理方案。

Best regards, LIN